

QuMOS: Using Quantum Computing to accelerate analysis of multi-omics dataset for Vaccine Development

Research Motivation

- The impact of quantum computing has begun to revolutionize the performance of algorithms, opening new possibilities into bioengineering.
- Machine learning algorithms can handle both regression and classification tasks, their performance gradually diminishes as the size of datasets increases. This decline in efficiency and scalability has become a critical issue that needs to be solved.
- Richard Feynman has inspired the challenge of prompting researchers to explore innovative approaches to enhance or replace existing machine learning methods with quantum mechanisms, enabling better classification and recognition in larger datasets.
- Quantum algorithms and simulations can model molecular interactions with a level of detail that classical computers cannot achieve, providing deeper insights into biological systems.

Introduction

The goal of this project is to develop and understand quantum algorithms for the processing of genomics dataset (e.g., NCBI SARS-CoV2 dataset) as applied to vaccine development. Through encoding its data into quantum circuits and gates across an algorithm which will search for specific genetic markers and patterns in a large dataset. Then implement quantum machine learning (QML) for mutation analysis in SARS-CoV2 spike protein dataset to identify and classify mutations in the SARS-CoV-2 spike protein with fidelity. Consequently, we will examine and create the quantum simulation for mutation analysis into a Quantum SVM which will map our mutation analysis data into a quantum feature space, using quantum kernels to map the data into qubits that can be analyzed through gates like Hadamard and Multi-Pauli Gates (ZZ FeatureMap). This will enable us to analyze whether quantum enhanced algorithms can truly accelerate multi-omics datasets.

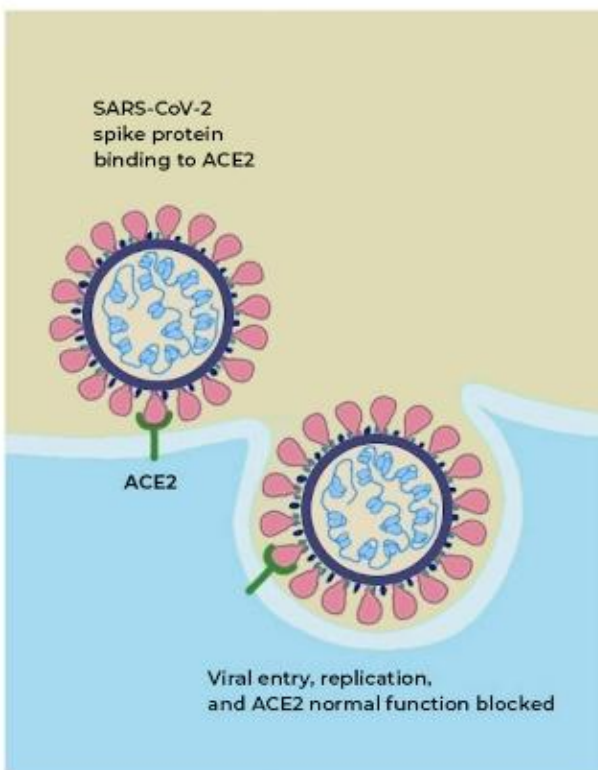
- Research Contributions**
- Previous studies have discovered theoretical algorithms to analyze different type of quantum gates or create them.
 - The promising potential of quantum computing, defined by Richard Feynman in 1982, has led researchers into capturing the intricate probabilistic nature of quantum phenomena in complex simulations or simple scenario
 - Useful for researchers to understand systems of electrons and the probabilities of their location

Research Objectives and Design

- To efficiently develop and integrate quantum algorithms for modeling the effects of quantum computing on accelerating vaccine development with high, efficient fidelity
- Build an system that transforms classical data storage into quantum states with exponential capacity to store genomic sequences of SARS-CoV-2 Spike Protein Dataset
- To create the idea of “*Quantum Machine Learning*” with utilizing quantum feature spaces and quantum computing techniques to supervised learning algorithms to classify genetic variations or improve vaccine development

Pre-Processing of the Spike Protein Dataset

The NCBI SARS-CoV-2 spike protein dataset is a specialized collection of data maintained by the National Center for Biotechnology Information (NCBI), focusing specifically on the spike protein of the SARS-CoV-2 virus in order to combat our COVID-19 Pandemic. This dataset specifically includes genetic sequences of the spike protein from various strains of the SARS-CoV-2 virus that can be crucial for identifying mutations



I. SPIKE PROTEIN'S ACE2 RECEPTOR ON HUMAN CELLS

NCBI SARS-CoV-2 dataset file structure is based on FASTA file format. A FASTA file is a simple, text-based format for representing nucleotide sequences (DNA or RNA) or amino acid sequences (proteins). For example, a Spike Protein FASTA file entry is shown below:

```
>YP_009724390.1 spike glycoprotein [Severe acute respiratory syndrome coronavirus 2]

MFVFLVLLPLVSSQCNVLTRTQLPAYTNSFTRGVVYPDKVFRS
SVLHSTQDLFLPFFSNVTWEHAIHVSQGTNGTKRFDPVLPFNQGV
YFASTEKSNIRGWIGFTTLDSTKQSLIVNNATNVVVKVCEQFQ
CNDPFLGVYHKNKSWMESEFRVYSANNCITFEYVVSQFLMDLE
GKQKFNRLREFFVFNKIDFVYSKPTPIINVRDLQGSFALAP
LVLDPIGIMITRTPTLLALHRSYLTFGDSSSGWFGAAAYVGYL
QPRTEFLKYNGTITDAVCDLPLSETKCTLKSTFEKGIYQT
SNQ...
```

Each letter in the sequence represents an amino acid, following the standard one-letter amino acid codes (e.g., M for Methionine, F for Phenylalanine). These types of sequences are critical for understanding how the virus binds to host cells and develops into more mutations. As part of our analysis of FASTA spike protein datasets, we need to look for genetic markers and epitopes within the spike protein sequence that can be recognized by the immune system to identify critical regions of the virus's infectivity, such as its receptor-binding domain to the ACE2 Receptor

Quantum Machine Learning for Mutation Analysis

To begin, **QML** is a variant of machine learning with focus on quantum properties like entanglement, superposition, and parallelism to enhance computational capabilities, particularly in the context of high-dimensional quantum representation spaces known as Hilbert spaces.

- **Superposition:** A qubit can represent both 0 and 1 at the same time, allowing quantum computers to process a vast amount of information simultaneously in a parallelistic manner.
- **Entanglement:** When qubits become entangled, the state of one qubit is directly related to the state of another, no matter how far apart they are.
- **Hilbert Spaces:** Spaces that enable efficient storage and manipulation of vast amounts of information through entanglement that machine learning cannot recreate.

$$\frac{1}{\sqrt{2}}(|0\rangle_A \otimes |1\rangle_B - |1\rangle_A \otimes |0\rangle_B)$$

By representing data in these quantum-enhanced spaces, QML models can process and analyze information that classical systems would find computationally prohibitive, paving the way for us to analyze and accelerate complex and expansive systems like SARS-CoV-2 Datasets with unparalleled identification.

In the context of genomic analysis and sequencing though, quantum parallelism and annealing can enhance classification tasks by exploiting entanglement and advanced operators like projection and unitary transformations.

- **Projection Operators:** Selectively focus on specific mutations (vectors) or sequences within a dataset
- **Unitary Operators:** Operating in Hilbert spaces, it transforms the quantum states for further analysis that would be beneficial for the qSVC and qSVM later:

$$\hat{\mathbb{P}}_{ab}|\psi\rangle = |a\rangle\langle a|\psi\rangle + |b\rangle\langle b|\psi\rangle = \alpha|a\rangle + \beta|b\rangle \quad \langle Ux, Uy \rangle_{H_2} = \langle x, y \rangle_{H_1} \quad \text{for all } x, y \in H_1.$$

The **computation of Fidelity** has a strong relation to the Quantum Approximate Optimization Algorithm (QAOA) and Fidelity Algorithms. In QAOA, fidelity evaluates how closely the quantum state approximates the desired ground state. QAOA is a hybrid quantum algorithm that alternates between two unitary operators (Ansatz Construction), preserving the length of quantum states.

V. COMPUTATION OF FIDELITY

$$F(\rho, \sigma) = \left(\text{Tr} \left[\sqrt{\sqrt{\rho}\sigma\sqrt{\rho}} \right] \right)^2$$

The impact of the fidelity equation in relation to QAOA shows that higher fidelity implies the QAOA state closely approximates the desired ground state which connects to “TR”, meaning the trace relation between two states (ρ & σ) of 0 and 1 to track fidelity, improving the algorithm’s effectiveness in evaluating genomic datasets. Fidelity algorithms also serve to quantify errors by measuring how much decoherence in a quantum system causes the prepared state to deviate from the target state

The computation of entanglement will utilize various feature mapping algorithms to integrate quantum machine learning (QML) for mutation analysis. Specifically, the ZZFeature Map will be applied to encode classical data into quantum states. These maps achieve this by employing rotations around the Z-axis using unitary operators

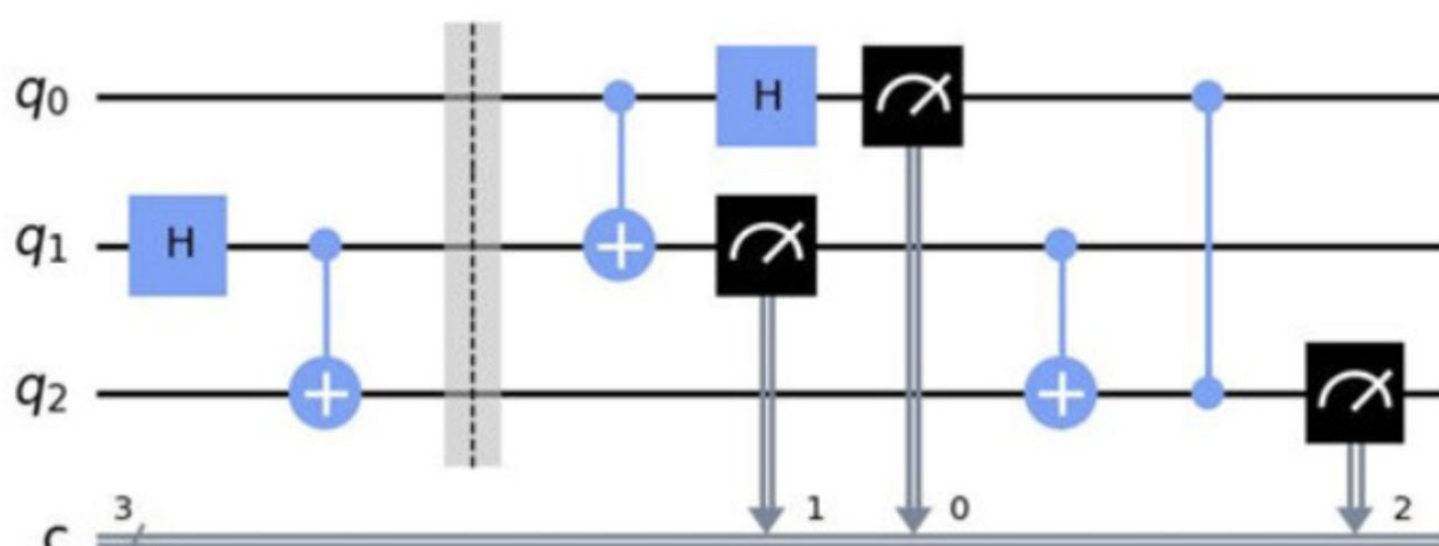
Implementation of Quantum Circuits and Gates of Mutation Analysis

To create a Quantum Circuit with multiple algorithms and gates that will be advantageous, it begins with encoding genomics data into quantum states so we can utilize the quantum superposition and entanglement in the nucleotide mapping. After creating and encoding the genomic data seen through use of looping and mapping through all nucleotides in the spike protein dataset

We will implement quantum gates into encoded quantum states for nucleotides using methods like the Hadamard Gate, SWAP Gate, and ZZFeature Map.

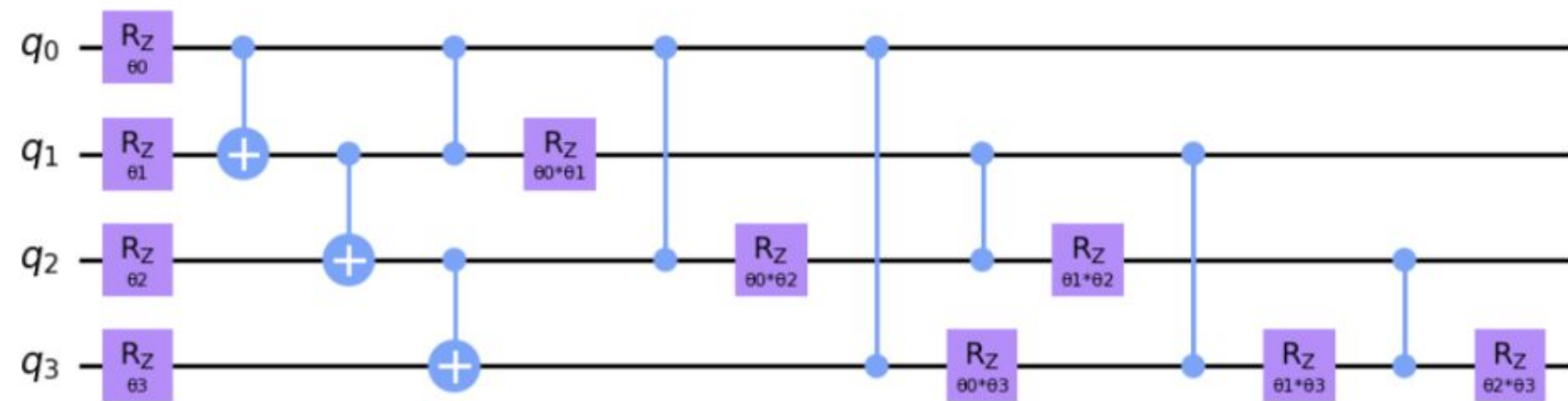
- **Hadamard Gate:** This gate creates superposition, enabling quantum states to explore multiple possibilities simultaneously between each other
- **SWAP Gate:** This gate changes quantum states between qubits, essential for entanglement and state manipulation
- **ZZ Feature Map:** This gate applies rotations around the Z-axis to represent classical data within the quantum domain.
- **Controlled-Z & X Gate:** This gate adds entanglement and phase shift that is conditional on their states of other qubits

II. QUANTUM CIRCUIT OF HADAMARD AND CONTROL-Z GATES



From all of these gates in the circuit, the encoded nucleotide sequences that were manipulated into qubits can now be supported with deeper analysis of correlations in genomic sequences, being able for us to track mutations. This implementation is attached above:

III. QUANTUM CIRCUIT OF ZZ FEATURE MAP



The impact of illustrating the ZZ Feature Map is that it transforms the genomic data into a quantum feature space where each analyzed nucleotide is encoded as a parameter into the R_z rotation gates into input data along the multi-dimensional space. As it allows for entanglement operation to connect correlation between the qubits, it can also leverage quantum parallelism across quantum feature space to represent all possible mutations

Quantum Support Vector Classifier and Machine Analysis

A qSVM/qSVC is a popular kernel-based method for classification, but what exactly is a kernel method? A kernel function measures the similarity between data points in the quantum feature space by leveraging entangled qubits and quantum transformations.

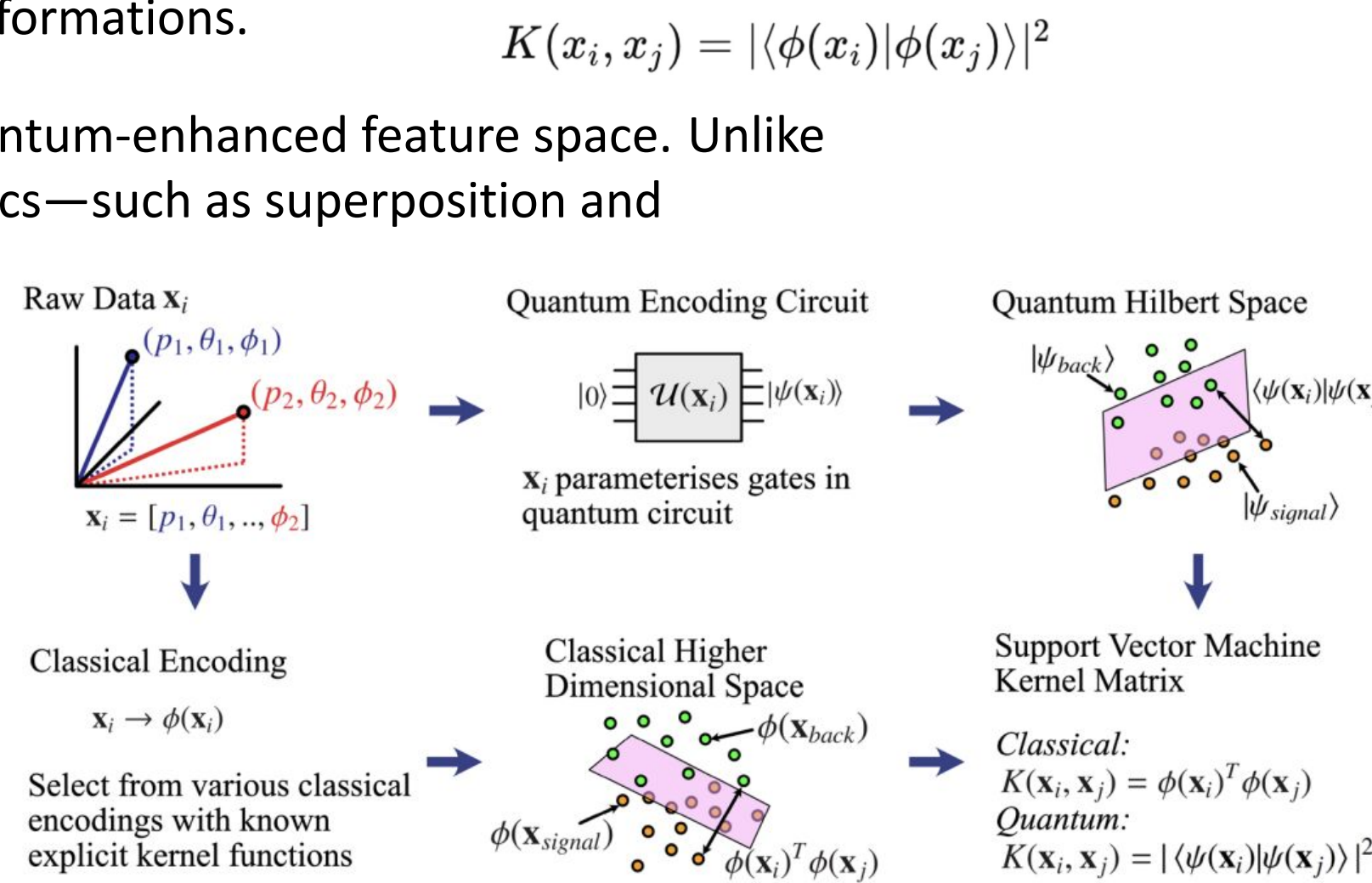
- This kernel matrix element captures the relationships between data points in the quantum-enhanced feature space. Unlike classical kernels, quantum kernels exploit the unique properties of quantum mechanics—such as superposition and entanglement—to represent non-linear patterns (mutations) more effectively.

In this context, quantum kernels use quantum feature maps, specifically the ZZ Feature Map, to calculate the kernel. The quantum feature map embeds classical nucleotide data into the quantum Hilbert space, enabling the qSVM to exploit quantum properties like entanglement and superposition.

- We can apply fidelity-based metrics to reveal how accurate the quantum feature map is to analyze the genomic sequences

In a quantum SVM, the quantum hyperplane represents the decision boundary in the high-dimensional quantum Hilbert space. This quantum hyperplane is particularly significant for genomic data analysis, as it can classify and map nucleotide sequences of the spike protein, identifying specific mutations.

These quantum states are entangled, and the quantum SVM leverages the encoded data to compute the inner product of quantum states, which forms the kernel estimation circuit.



IV. The Encoding Process of Quantum SVM

Kernel Estimation Circuit

Now, the classification of nucleotide sequences based on their similarities in the quantum feature space may become entangled in the Hilbert Space, we need to test fidelity between the original and current quantum states of the qubits after running them through the genomic sequence of markers and the main quantum circuits.

- To test Fidelity, the **Kernel Estimation Circuit** can be repeated iteratively for combinations and mutations of encoded quantum nucleotides until a complete quantum kernel matrix is obtained which represent the classification of nucleotide sequences based on their similarities into quantum hyperplane to separate clusters of nucleotide sequences and detect mutations with high precision

Creating Grover’s Algorithm for Genomic Sequencing

At last, we know that Quantum Computing are capable of highly, intensive algorithms, but what makes it truly different from classical computing? It’s exponential speedup to accelerate vaccine development through Grover’s Algorithm, the true goal of this project.

Grover’s Algorithm is a quantum algorithm that provides a quadratic speedup for unstructured search problems. An unstructured search is like looking for a specific item in an unsorted database or list. Classical algorithms require, on average, $O(N/2)$ checks for N items, while Grover’s algorithm can find the solution in roughly $O(\sqrt{N})$ steps. This makes it significantly faster for large datasets, although the speedup is quadratic rather than exponential through the use of oracle algorithm, which means it uses unitary operators to augment time complexity through CZ Gate and Hadamard.

To create a quantum algorithm for genomic sequencing, we utilize the **variational quantum eigensolver (VQE)** to analyze the binary representation of genetic markers as quantum states and perform specific operations on them after processing through their respective circuits. In this successful implementation, **Grover’s algorithm is applied to identify the specific genetic mutation sample 1010 from a set of genetic markers, including 1101, 1010, 1111, 0000.**

```
Set up a qSVM to classify two groups of data,
representing a # model of "mutations" in the spike
protein

# Generate a synthetic dataset (placeholder for real
genomic
# data)
features, labels = make_circles(n_samples=200,
noise=0.1, factor=0.2, random_state=123)
labels = labels * 2 - 1 # Convert labels to -1 or 1
for qsvm
# Encode classical data to a quantum state
from qiskit.circuit.library import ZZFeatureMap

feature_dim = 2 # Assuming 2 features for simplicity
feature_map =
ZZFeatureMap(feature_dimension=feature_dim, reps=2)

# Setup the qSVM model and train it

quantum_kernel =
QuantumKernel(feature_map=feature_map,
quantum_instance=QuantumInstance(Aer.get_backend('sta
tevector_simulator'))

qsvm = qSVM(quantum_kernel)

# Now make predictions
predicted_labels = qsvm.predict(features)

# Evaluate the model

Example to use Grover's algorithm to find the binary-encoded
"genetic marker" 1010 in a "dataset" of four markers: 1000

# Oracle for the marker '1000'
def oracle(circuit):
    circuit.cx(0, 1) # Example of an oracle
    operation
    circuit.cx(2, 3)

# Initialize circuit with 4 qubits
n = 4
grover_circuit = QuantumCircuit(n, n)

# Apply Hadamard gates
grover_circuit.h(range(n))

# Apply the oracle
oracle(grover_circuit)

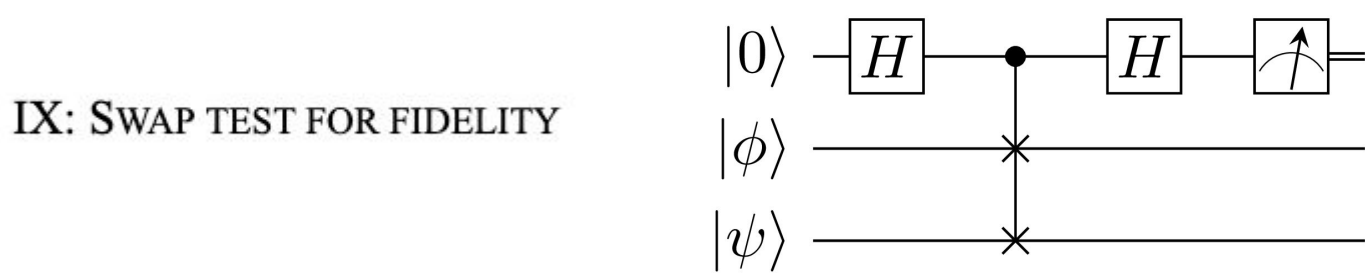
# Apply the diffuser
# The diffuser is a key part of Grover's algorithm
that amplifies the amplitude of the marked
state(s).
grover_circuit.append(diffuser(n), range(n))

# Measurement
grover_circuit.measure(range(n), range(n))

# Execute the circuit
aer_sim = Aer.get_backend('aer_simulator')
transpiled_grover_circuit =
transpile(grover_circuit, aer_sim)
results =
aer_sim.run(transpiled_grover_circuit).result()
answer = results.get_counts()

# Plot the results
plot_histogram(answer)
plt.show()
```

Lastly, fidelity testing is performed using a **SWAP gate** test, which exchanges the quantum states of two qubits to ensure consistency and entanglement between states. This test ensures that even after applying Grover’s algorithm and additional gates, the encoded quantum states retain their integrity, which is crucial for accurate genomic sequencing and mutation analysis.



The use of the SWAP test was utilized to measure the fidelity between the original and current quantum states of the qubits after running them through the genomic sequence of markers. Using the imported qSVM model, which was trained for classification, the final quantum state demonstrated a high degree of similarity. These results highlight the effectiveness of quantum algorithms in accelerating multi-omics data analysis, specifically with the spike protein dataset for binary classification tasks.

Discussion and Future Work

The primary goal of this research was to develop quantum algorithms for processing genomic datasets, particularly focusing on the SARS-CoV-2 spike protein dataset. Our results demonstrated that encoding genomic data into quantum states allowed for accurate classification and analysis of mutations in the **SARS-CoV-2 spike protein**. The **qSVM model**, enhanced with **quantum kernel methods** and **ZZ Feature Map**, deeper insights with using **superposition**, **entanglement**, and **quantum parallelism**. Grover’s algorithm, integrated with our circuits, efficiently identified specific genetic markers in a dataset, while the **Variational Quantum Eigensolver (VQE)** and **Fidelity’s Algorithm** ensured high accuracy and robustness of our quantum operations.

- That said, QuMOS is a mere proof of the capabilities that quantum computing can bring in the future for the power of analyzing multi-omic datasets. I did successfully (one of first people) connect the thoughts of qSVM to acceleration of vaccine development. In the future, **QuMOS can truly be implemented with the substantial rise in quantum computing areas such as recent innovations like the Majorana 1 Chip.**

Key References

Grover, L.K.: A fast quantum mechanical algorithm for database search. <i>ACM. 212-219</i> (1996)	Heredge, Jamie & Hill, Charles & Hollenberg, Lloyd & Seavor, Martin. (2021). Quantum Support Vector Machines for Continuum Suppression in B Meson Decays. <i>Computing and Software for Big Science</i> . 10.1007/s41781-021-00075-x.	Patrick Rebertost, Masoud Mohseni, and Seth Lloyd. Quantum support vector machine for big feature and big data classification. <i>arXiv preprint arXiv:1307.0471</i> , 2013.
Modeling Quantum Machine Learning for Genomic Data Analysis: Navneet Singh, Shiva Raj Pokhrel, IEEE		All Other References are Printed on Sheet